KAMI: A Knowledge-driven Artificial Intelligence Platform for Predictive Water Governance, Infrastructure Intelligence, and Smart Resource Management

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Abstract

Water scarcity, aging infrastructure, rapid urbanization, climate change, and fragmented public service delivery continue to challenge sustainable water management. Existing digital water management solutions largely focus on information dissemination and complaint registration, offering limited predictive intelligence, automation, and decision support. KAMI (Knowledge-driven Artificial Intelligence for Mitigating Water Insecurity) addresses these limitations through a unified, AI-powered platform that combines Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), multimodal AI, computer vision, geospatial intelligence, IoT sensor networks, predictive analytics, reinforcement learning, optimization algorithms, and digital twin technology. The platform enables multilingual citizen interaction through text, voice, images, videos, documents, and location-aware reporting while automatically detecting leaks, contamination, damaged infrastructure, flooding, and illegal water connections using computer vision. Predictive machine learning models forecast water demand, groundwater availability, reservoir depletion, droughts, floods, equipment failures, and maintenance requirements using historical and real-time data. Optimization algorithms improve tanker routing, pump scheduling, valve operations, and maintenance planning, while explainable AI provides transparent recommendations supported by confidence scores and evidence. Interactive dashboards with real-time heatmaps, anomaly detection, and infrastructure health monitoring empower administrators to make proactive, data-driven decisions. KAMI offers a scalable and extensible framework capable of improving water governance today while providing a strong foundation for future smart-city applications in sanitation, electricity, transportation, healthcare, agriculture, and disaster management.